KB금융그룹 | 국민의 평생 금융파트너

K-디지털 트레이닝

스프링과 MySQL Database 연동

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데이터베이스 생성 및 계정 생성, 권한 설정

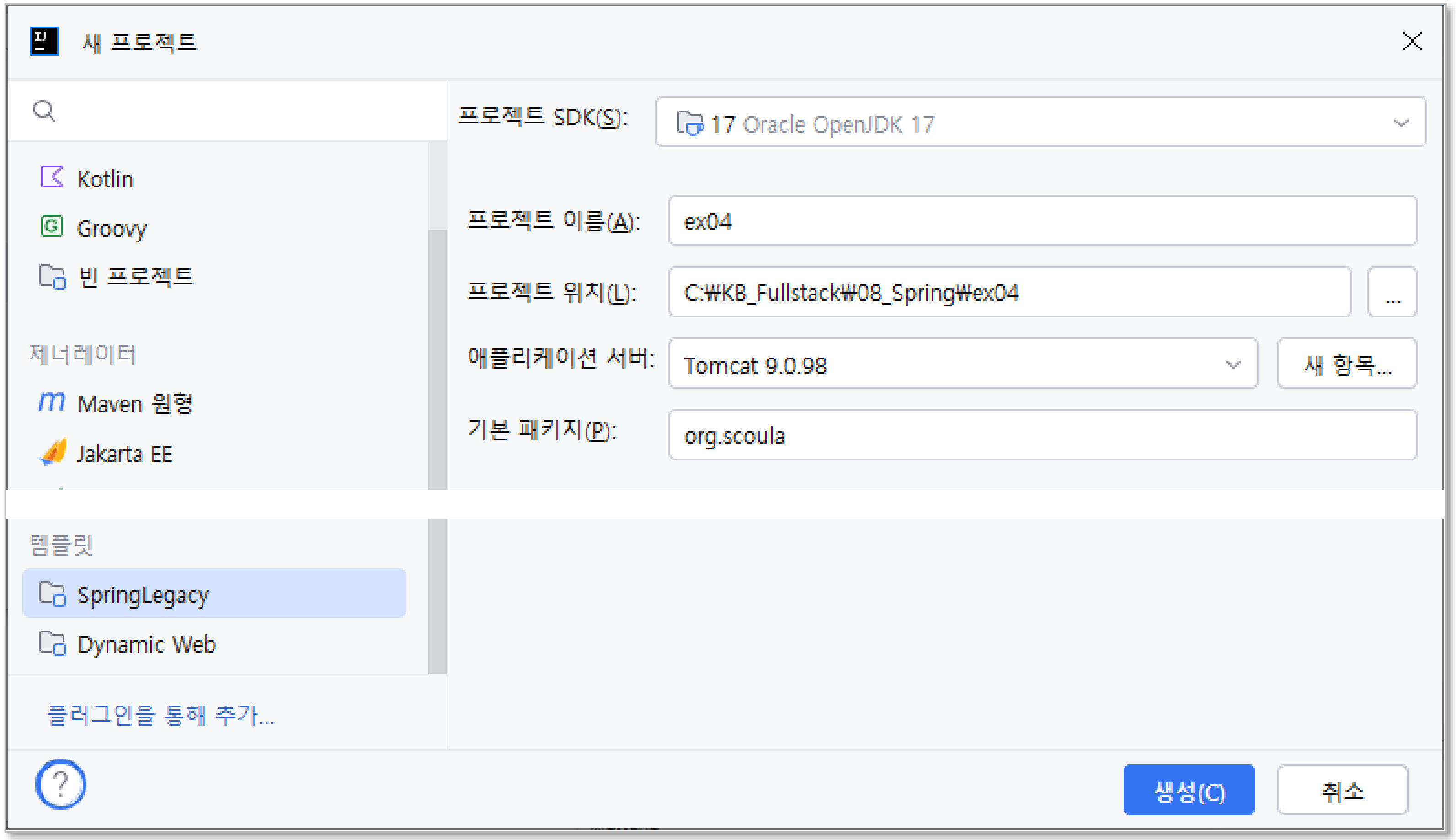
```
create database scoula_db;

create user 'scoula'@'%' identified by '1234';

grant all privileges on scoula_db.* to 'scoula'@'%';
```

프로젝트 만들기

- Templates: SpringLegacy
- Project name: ex04

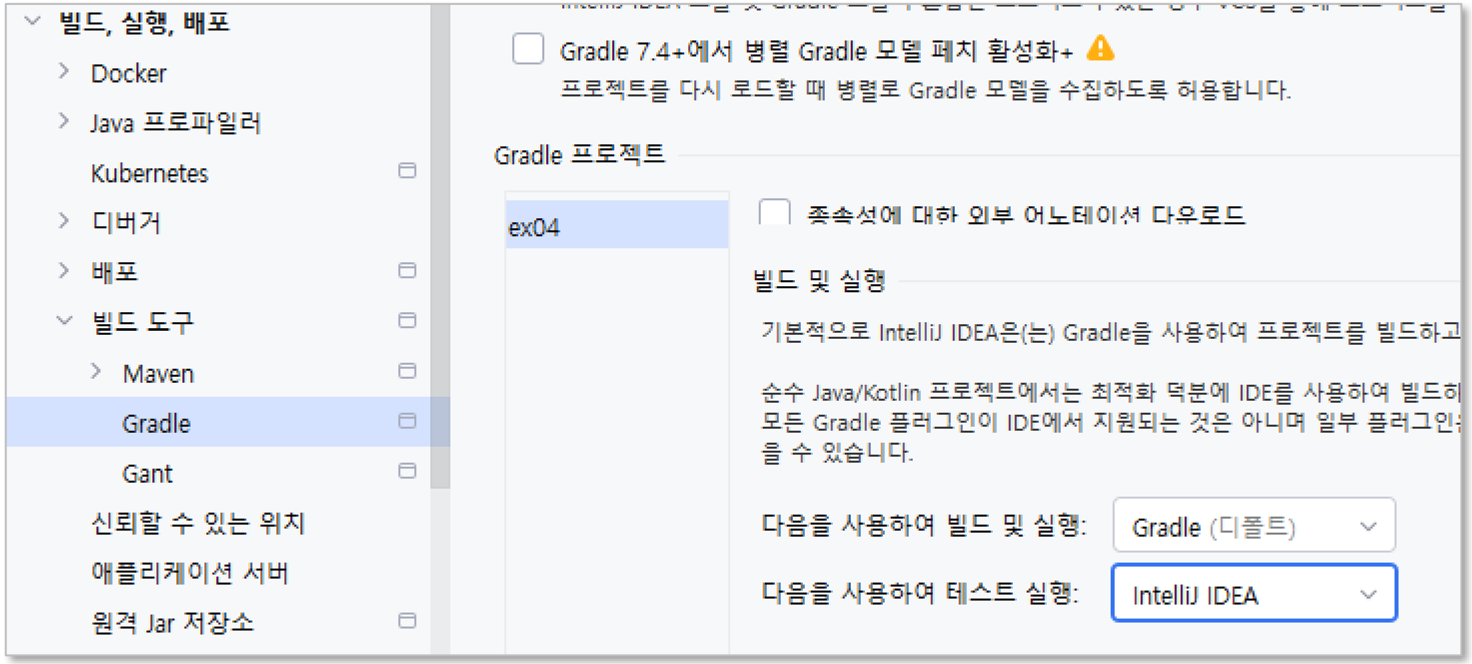


settings.gradle

```
rootProject.name = 'ex04'
```

설정

- Enable Annotation Processor 활성화
- 빌드, 실행, 배포 > 빌드 도구 > Gradle
  - 다음을 사용하여 테스트 실행: IntelliJ IDEA 선택



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프로젝트의 JDBC 연결

build.gradle

```
// 데이터베이스
implementation 'com.mysql:mysql-connector-j:8.1.0'
```

→ gradle sync

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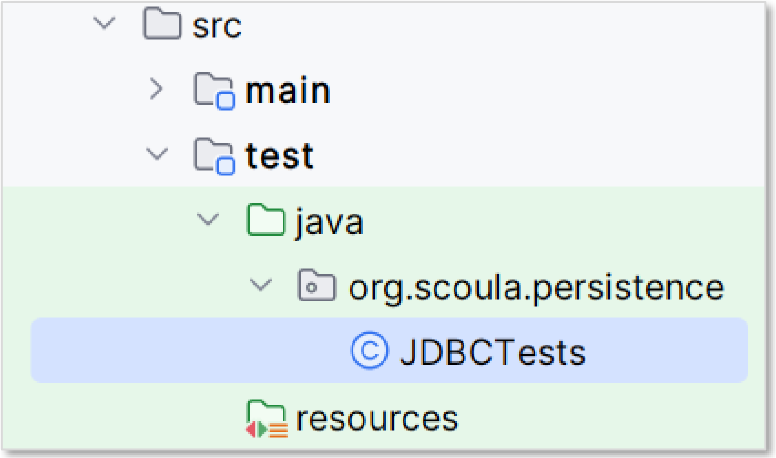
JDBC 테스트 코드

- src/test/java
  - org.scoula.persistence
    - JDBCTests.java

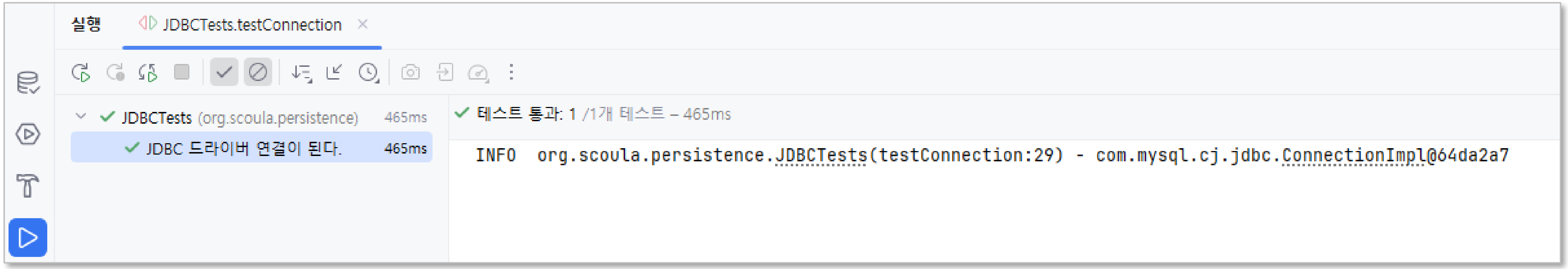


test :: JDBCTest.java

```
package org.scoula.persistence;

import lombok.extern.log4j.Log4j2;
import org.junit.jupiter.api.BeforeAll;
import org.junit.jupiter.api.DisplayName;
import org.junit.jupiter.api.Test;

import java.sql.Connection;
import java.sql.DriverManager;
import static org.junit.jupiter.api.Assertions.fail;

@Log4j2
public class JDBCTest {
    @BeforeAll
    public static void setup() {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch(Exception e) {
            e.printStackTrace();
        }
    }
```

test :: JDBCTests.java

```java
  @Test
  @DisplayName("JDBC 드라이버 연결이 된다.")
  public void testConnection() {
    String url = "jdbc:mysql://localhost:3306/scoula_db";
    try(Connection con = DriverManager.getConnection(url, "scoula", "1234")) {
      log.info(con);
    } catch(Exception e) {
      fail(e.getMessage());
    }
  }

}
```



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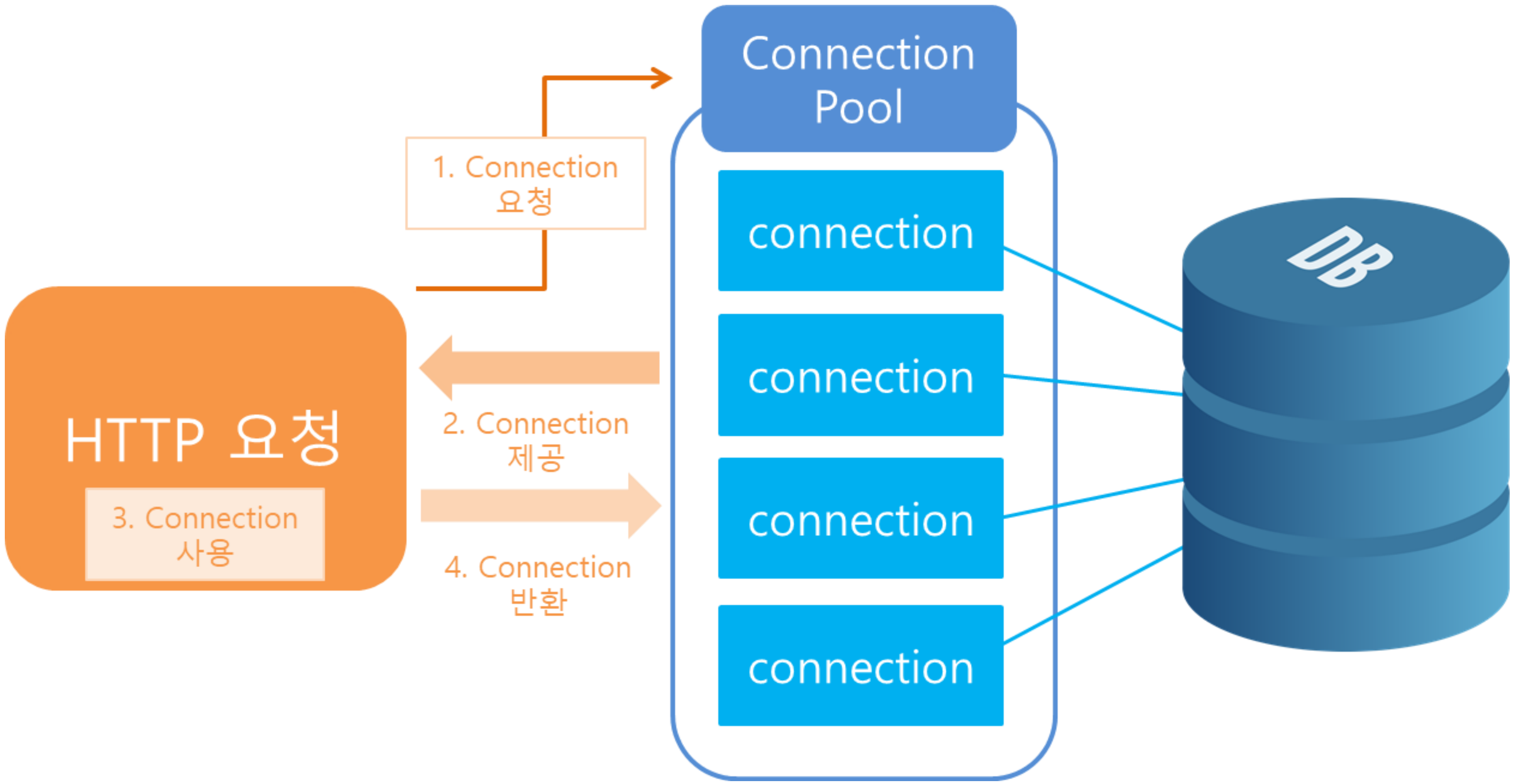
DataSource

- Connection Pool



라이브러리 추가와 DataSource 설정

build.gradle

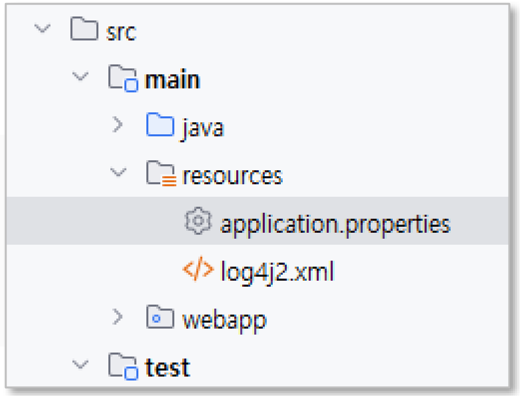
```
// 데이터베이스
implementation 'com.mysql:mysql-connector-j:8.1.0'
implementation 'com.zaxxer:HikariCP:2.7.4'
```

→ gradle sync

4 커넥션 풀 설정

resources/application.properties

```
jdbc.driver=com.mysql.cj.jdbc.Driver
jdbc.url=jdbc:mysql://127.0.0.1:3306/scoula_db
jdbc.username=scoula
jdbc.password=1234
```



.properties에서 값 읽어 오기

- @PropertySource({"properties 경로 문자열"})
  - 클래스 레벨 어노테이션
  - 사용할 .properties 경로를 지정
  - 예
    @PropertySource({"classpath:/application.properties"})

- @Value("${키:기본값}")
  - 필드 레벨 어노테이션
  - 기본값은 생략 가능
  - 예
    @Value("${jdbc.driver}") String driver;

RootConfig.java

```
package org.scoula.config;

import com.zaxxer.hikari.HikariConfig;
import com.zaxxer.hikari.HikariDataSource;
import org.apache.ibatis.session.SqlSessionFactory;
import org.mybatis.spring.SqlSessionFactoryBean;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.beans.factory.annotation.Value;
import org.springframework.context.ApplicationContext;
import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.context.annotation.PropertySource;
import org.springframework.jdbc.datasource.DataSourceTransactionManager;

import javax.sql.DataSource;

@Configuration
@PropertySource({"classpath:/application.properties"})
public class RootConfig {
    @Value("${jdbc.driver}") String driver;
    @Value("${jdbc.url}") String url;
    @Value("${jdbc.username}") String username;
    @Value("${jdbc.password}") String password;
```

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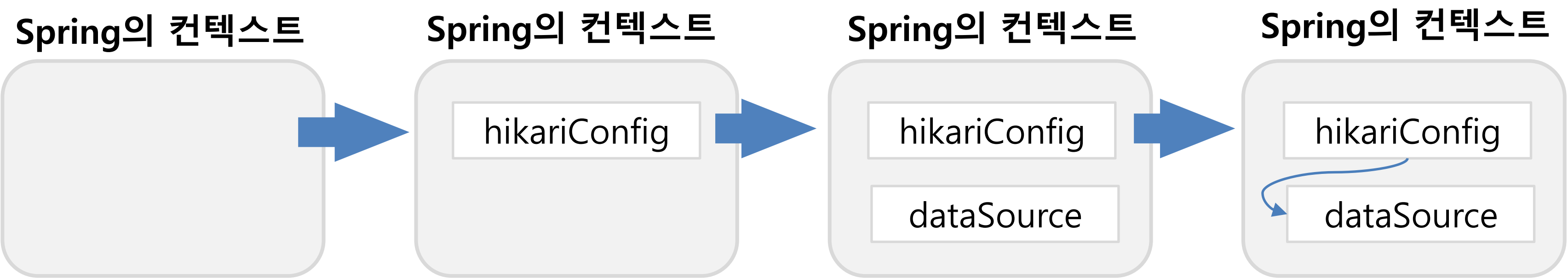
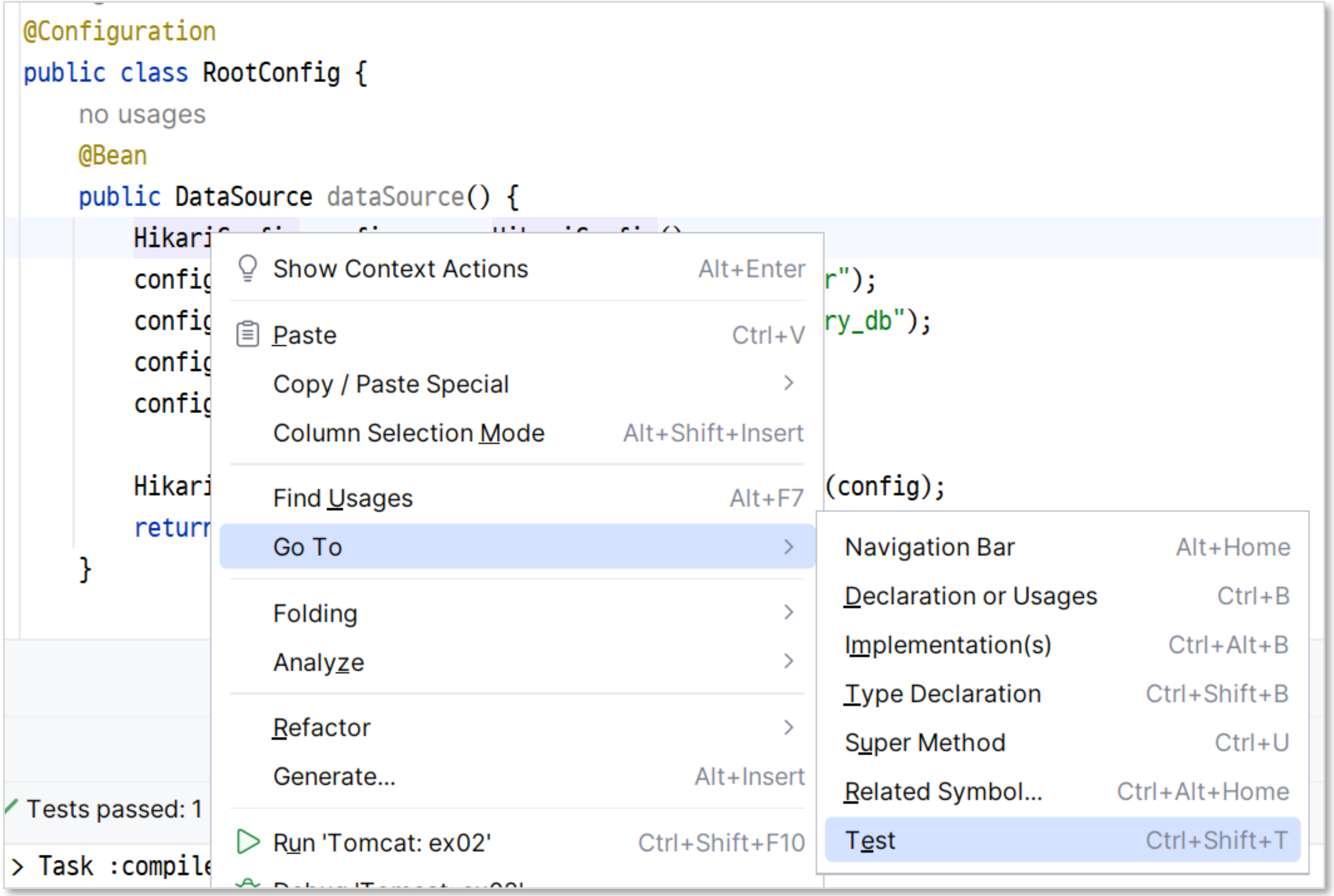
RootConfig.java

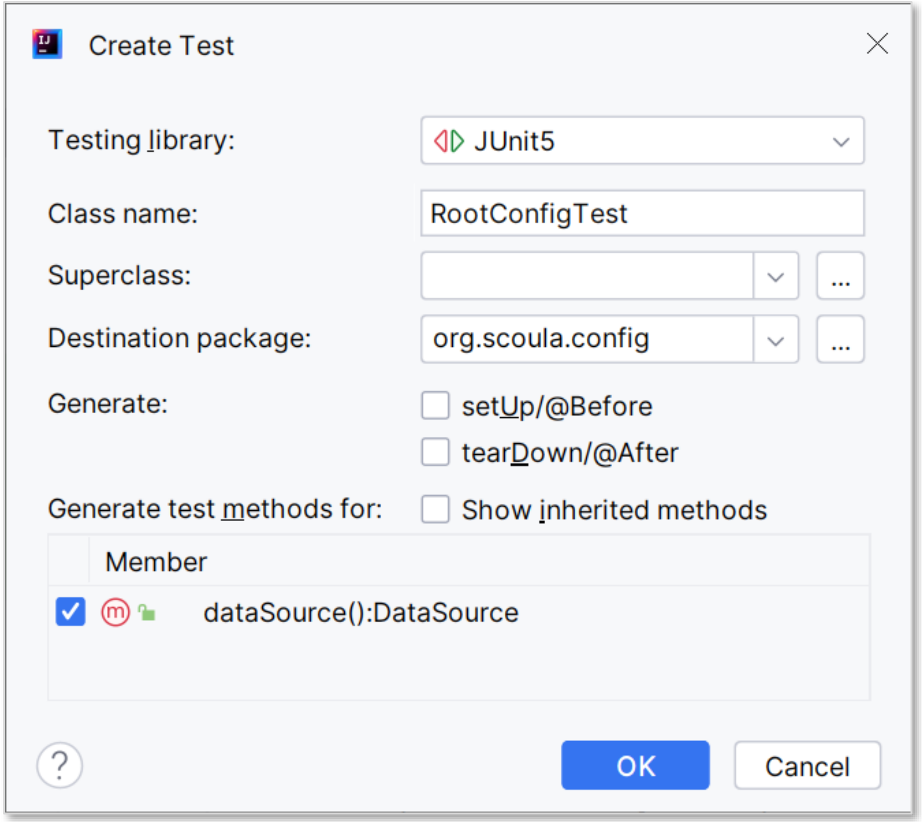
```
    @Bean
    public DataSource dataSource() {
        HikariConfig config = new HikariConfig();

        config.setDriverClassName(driver);
        config.setJdbcUrl(url);
        config.setUsername(username);
        config.setPassword(password);

        HikariDataSource dataSource = new HikariDataSource(config);
        return dataSource;
    }
}
```

라이브러리 추가와 DataSource 설정



RootConfig 테스트 클래스 만들기





test :: config.DataSourceTests.java

```java
@ExtendWith(SpringExtension.class)
@ContextConfiguration(classes= {RootConfig.class})
@Log4j2
class RootConfigTest {

    @Autowired
    private DataSource dataSource;

    @Test
    @DisplayName("DataSource 연결이 된다.")
    public void dataSource() throws SQLException {
        try(Connection con = dataSource.getConnection()){
            log.info("DataSource 준비 완료");
            log.info(con);
        }
    }
}
```